Examiner only

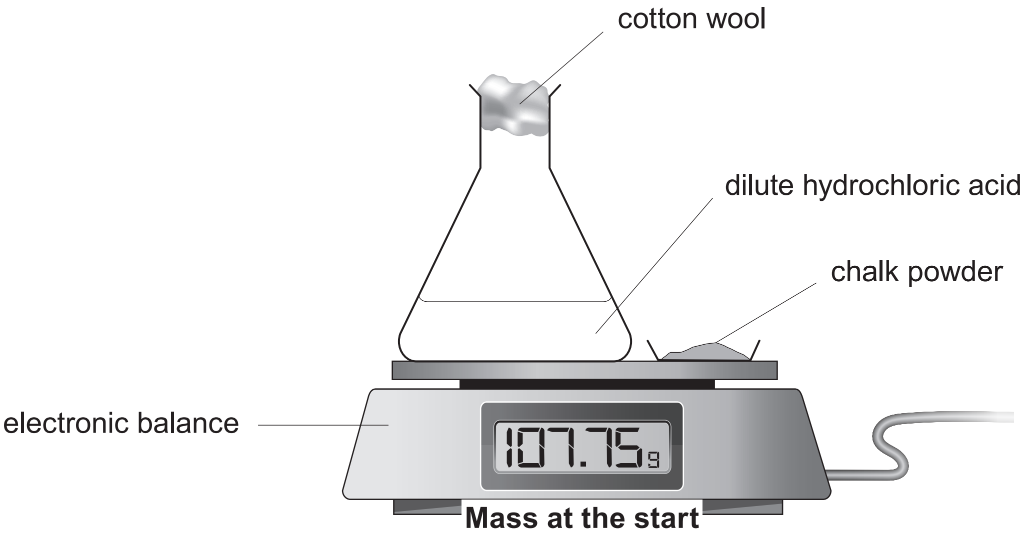
7. Chalk is a form of calcium carbonate, $CaCO_3$.

A student carried out an experiment to investigate the rate of reaction between **powdered** chalk and **excess** dilute hydrochloric acid at 20 °C.

Chalk reacts with dilute hydrochloric acid forming carbon dioxide gas.

$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

The diagram shows the apparatus used.



(a) State the purpose of the cotton wool. [1]

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(b) The student recorded the mass of the flask and contents every 10 seconds for 70 seconds.

Describe what the student must do to calculate the mass of carbon dioxide formed in the first 10 seconds. [1]

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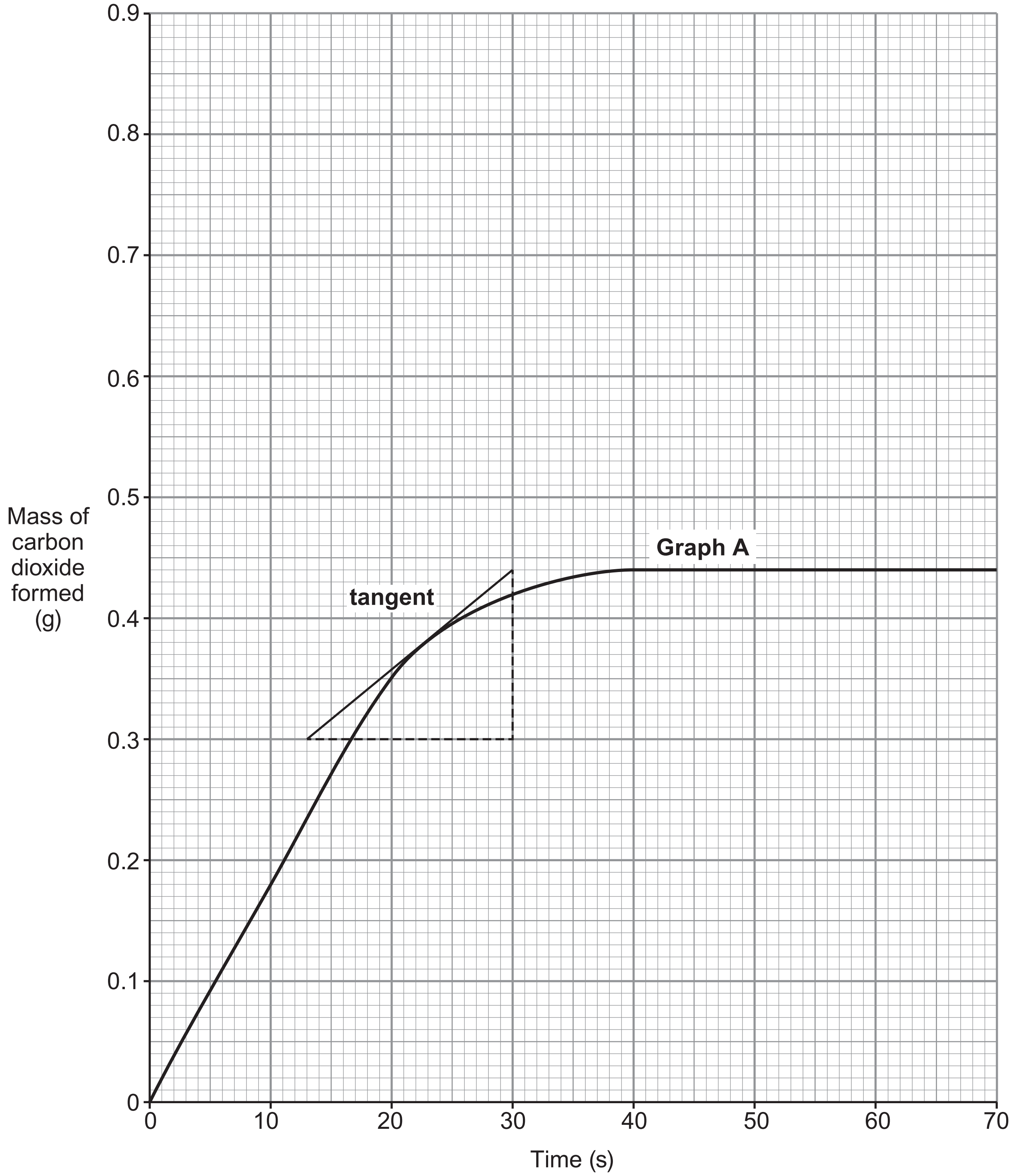
(c) Graph **A** shows the mass of carbon dioxide formed against time.

A tangent to the graph has been drawn at 23 s.



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Use the tangent and the equation below to calculate the rate of reaction at 23 s.

Show your working. Give your answer to **two** significant figures. [3]

$$\text{rate} = \frac{\text{change in mass}}{\text{change in time}}$$

Rate = ... g/s

(d) The student repeated the experiment using the **same** mass of chalk powder as the original experiment but a **different concentration** of acid. The table shows the mass of carbon dioxide formed.

Time (s)	0	10	20	30	40	50	60	70
Mass of carbon dioxide formed (g)	0.00	0.11	0.20	0.28	0.35	0.40	0.43	0.44

(i) Plot the results from the table on the grid opposite and draw a suitable line.

Label this graph **B**. [3]

(ii) Use particle theory to explain why the rate of this reaction is lower than that in the original experiment. [3]

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(e) Sketch on the grid the curve you would expect if the experiment was repeated using the **original** acid and **twice** the mass of chalk. Assume that the acid is still in excess.

Label this graph **C**. [1]

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